Generating Animal Hybrids Using Adversarial Autoencoders

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***Abstract*—Although the field of image generation has progressed significantly, the ability to combine images together to create reasonable mixtures has mostly been neglected. Previous advancements in a similar vein like Style Transfer [11] were restrictive in the specific traits one could combine (e.g. art style) and did not allow for the mixing of content. In this report, we describe advancements in image interpolation and focus on recreating results of ACAI and AEAI models to show the capabilities of adversarial autoencoder techniques for image interpolation [2], [15]. We test on animal faces, and show that AEAI beats ACAI in qualitative and quantitative tests. In most cases, AEAI produces realistic and believable animal hybrids.**

***Index Terms*—Autoencoders, Generative Adversarial Networks, Image Interpolation, Convolutional Neural Networks**

I. Introduction

In recent years, the field of image generation has developed very quickly; technologies such as Stable Diffusion and Imagen are able to create believable images from text prompts, and techniques for Style Transfer can change an image according to style prompts. However the ability to combine images together has been relatively neglected. By creating believable animal fusions, we plan to demonstrate the capacity of deep learning models to learn the core features of an image and also increase the creative freedom users have in image generation. Creating quality fusions can also be applied to data augmentation as a way to improve the generalization of machine learning models.

A trivial method of merging images together would be direct linear interpolation; in other words, a simple cross-fade between them. However, this almost always creates unbelievable results. A more reasonable approach is to first use an autoencoder to compress the images into a lower dimensional latent space, and then interpolate there. This would ideally capture and mix more abstract features of the image. This approach is still somewhat flawed, as the latent space might not be convex (i.e. the linear interpolation might not be a valid image), and can also lead to blurry images.

A solution that has been recently explored is the introduction of an adversarial component to the model which encourages the mixture of latents to produce a believable image. This strategy combines some aspects of the autoencoder (AE) and the Generative Adversarial Network (GAN), where the generator is the autoencoder and mixture of latents, and the discriminator pushes the generator to create more realistic fusions. Many variations of this AE + GAN architecture have been researched over the years, and the effectiveness is discussed in more detail in the literature review.

Another possibility is to utilize recent advancements in diffusion based models, which work similarly to traditional autoencoders but have a fixed, untrainable encoder that simply adds noise to the input. Interpolation in this space has been shown to be somewhat effective at generating reasonable interpolations, as seen in [1] of the literature review. In this paper, we show the capabilities of adversarial autoencoders for image interpolation, specifically for animal hybrid generation. We recreate and compare the results of the basic crossfade, ACAI, and AEAI.

II. Literature Review

There has been different strategies to tackle image interpolation, and image generation.

Image Generation. Diffusion models have been proven to be successful at image generation. Wang C. and Golland P. [1] interpolates between images using the latent space of diffusion models. They interpolate the latent images at varying levels of noise, and condition on context inputs, allowing for user controlled outputs.

Similar to [1], Rombach R. et al. [10] illustrate the capabilities of latent diffusion models for high quality and computationally efficient image generation. However, diffusion models require lots of data and training resources, which is beyond the scope of this project.

PixelVAE [7] illustrates the capabilities of variational autoencoders (VAE) for image generation by combining a VAE and PixelCNN.

disCVAE present a way generate images from user specified attributes (i.e. a text description) using a layered generative model (separating foreground and background) with disentangled latent variables learned from a VAE [9]. They present the usefulness of a disentangled latent space, and an alternative path to image interpolation.

Autoencoders for Image Interpolation. Our base model recreates ACAI, introduced in a seminal paper illustrating autoencoders ability to interpolate images [2]. ACAI combines latents of an image at varying degrees (adjusted through α parameter. The autoencoder is adversarially trained by a critic that tries to predict alpha.

Li G. et al. [3] builds upon the ACAI model by sampling alpha for each latent dimension, instead of one alpha for the

whole latent image. In addition, they introduce a correction module to move the interpolated image closer to the real latent space, in an attempt to produce more realistic interpolations.

Laine S. [5] suggests an improvement/alternative to ACAI's linear interpolation of images. They use a similarity metric derived from VGG-19 [14].

GAIA improves the classic AE/GAN architecture by using an autoencoder for both the generator and discriminator [6] (typically the discriminator is a CNN). They produce realistic images by staying withing the latent space of the real data distribution.

Christoper B. et. al builds upon ACAI and GAIA models by including augmented data into its training dataset, and doing crossover interpolation instead of linear interpolation [8]. Inspired by biology, crossover interpolation involves combining random features of latent with the remaining features from the other latent image.

StyleALAE combines an adversarial latent autoencoder with a StyleGAN generator, showing impressive results [4]. The latent autoencoder calculates reconstruction loss (reciprocity) in the latent space, rather than the image space. They show there model produces less entangled latent spaces.

Our final model is based on AEAI (Autoencoder Adversarial Interpolation) improves upon the ACAI model by using a real/interpolated image discriminator instead of a critic, and adds loss terms to the autoencoder [15]. The additional losses are cycle consistency and smoothness loss which encourages smoother interpolations. AEAI is able to keep smooth interpolations while creating more realistic images than the base ACAI model.

III. Problem Statement and Dataset

The goal of our model is to create believable and realistic fusions of pairs of animal images. This solution aims to address this task by utilizing advancements in autoencoders, image interpolation, and generative adversarial networks. This project will illustrate the possibilities in data augmentation, and creativity in image generation.

We interpolate on high quality images of animal faces, using the AFHQ dataset [18] with 16,130 colour images of size 512x512. The images contain three classes of animals; cats, dogs, and wild (canines and felines). We train on 14630 images and use 1500 for validation. The images were originally created for image synthesis, but we find that many of the traits desirable to generation are also desirable for interpolation. For instance, all images have consistent alignment and cropping, and relatively simple poses of animal faces.

IV. System Model

A. Pre-Processing

The original AFHQ dataset is downsampled using bilinear interpolation to a size of 128x128 to allow for faster training. The images are normalized to have mean 0 and standard deviation of 1 for each colour channel. This is also done to speed up training. The images are processed and normalized using the OpenCV library and are split into a train/validation sets.

B. ACAI Model

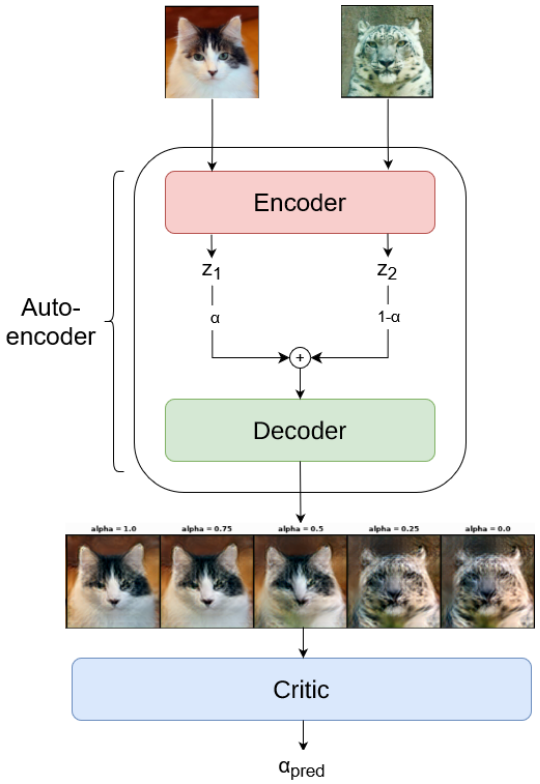


Fig. 1: Architecture diagram of the ACAI model.

The Adversarially Constrained Autoencoder Interpolation (ACAI) model is selected as the baseline for this project. This model was designed specifically for image interpolation and has shown success in a variety of datasets like MNIST. [2] It contains two significant components, an autoencoder and a critic.

1) Autoencoder: An autoencoder is a type of artificial neural network used to do feature extraction of images. It consists of two significant components: the encoder and decoder. The encoder half of the network is based on the VGG classifier model [**?**], and contains repeated convolution, ReLU activation and Max-pooling layers that result in a feature map before interpolation with another input image. The particular structure of the encoder is visualized in Fig. 3. The processed input first passes through a series of convolutional layers with a kernel size of 3x3 followed by ReLU activation and max pooling. The number of filters in these layers increases: starting from 64, moving to 128, and finally to 256. In the 256-filter stage, the convolution operation is applied twice before proceeding to the max pool. Following the convolutional layers, the output is flattened, converting the multi-dimensional feature maps into a one-dimensional vector. This vector is then fed into a fully connected layer, also equipped with ReLU activation. The decoder section attempts to reconstruct the interpolation

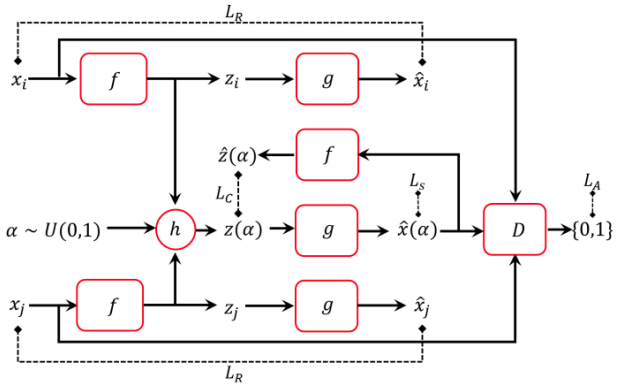


Fig. 2: AEAI original architecture diagram. The autoencoder consists of f (encoder) and g (decoder). The discriminator is D.

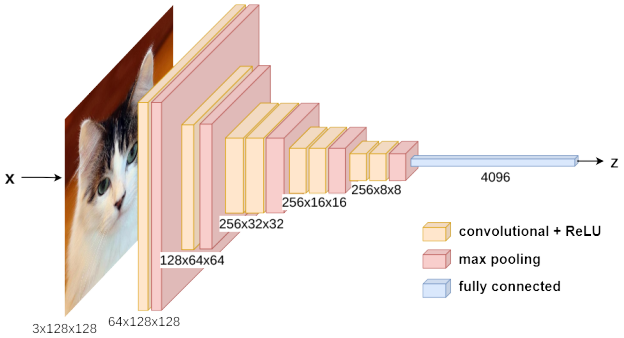


Fig. 3: Modified ACAI/AEAI Encoder based on VGG.

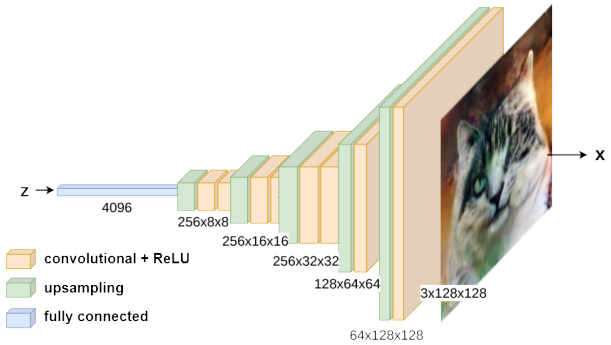


Fig. 4: Modified ACAI/AEAI Decoder based on VGG.

feature map by working on the reverse process of encoder. As a result, the Autoencoder is able to generate hybrid animal images.

2) Critic: The critic is somewhat similar to the discriminator part of GANs; it tries to differentiate the interpolated animal images from real images by generating the predicted value of interpolation factor. The architecture consists of same layer as the encoder in the autoencoder but with no repeated convolution filter in the 256 stage (see Fig. 5), as well as an additional two linear layers to project the output down to a single value. This structure is designed to intensively process and extract features from the output data of Autoencoder, providing a robust mechanism for the critic to evaluate the authenticity of the hybrid animal images.

C. AEAI Model

The Autoencoder Adversarial Interpolation model (AEAI) is the final model, which shows significant qualitative improvement from the classic ACAI model. This can be attributed to their goal of creating *admissible interpolations*. They define this by four conditions; boundary conditions, monotonicity, smoothness, and credibility. These conditions are satisfied by their proposed interpolation regularization mechanism, consisting of four loss components (see Loss Function section). Although these losses promote admissible interpolations, their ablation study reveals that the fourth component smoothness loss does not have significant qualitative impacts to the quality of image interpolations. For this reason, we chose to not include smoothness loss to our model.

1) Architecture: The main architecture of our model closely follows the paper, with slight changes to the autoencoder to suit our training capabilities and dataset (see Fig. 2). Similar to ACAI, we pass two images x_i and x_j through encoder f and linearly interpolate them with a sampled α (h in Fig. 2). The reconstruction loss L_R is calculated using a perceptual similarity loss [14], comparing the decoded image $\hat{x_j} = g(z_j)$, where g is the decoder, to the original image. This helps promote an effective latent representation and also ensures the model is capable of using the latent representation to regenerate the image. The latent interpolation $z(\alpha)$ is then decoded to produce $\hat{x}(\alpha) = g(z(\alpha))$, which is passed through the discriminator D along with a real image. The discriminator is then trained as a classifier which tries to distinguish between interpolated images and real images. It then predicts the probability that the interpolated image is real, which is used to penalize the autoencoder if it produces unrealistic images using the adversarial loss component (L_A). This is the negative log of the discriminator probability, which is the same as the binary cross entropy loss for a positive target. The cycle loss is then computed as the difference between the latent interpolation and the latent interpolation passed through decoder and encoder again; this helps ensure that the interpolated latents are reasonable in the learned latent space. Finally, if the smoothness loss were used, it would be computed as the norm of the gradient of the interpolated image with respect to α. This would ideally promote a smooth interpolation and helps avoid sudden jumps as α changes, but since it was found to be relatively ineffective in both our and the author's experiments, it was removed.

The autoencoder has the same architecture as the one we used in ACAI; this was to try and ensure that any differences in results were due to the improvements in AEAI and not the architecture itself.

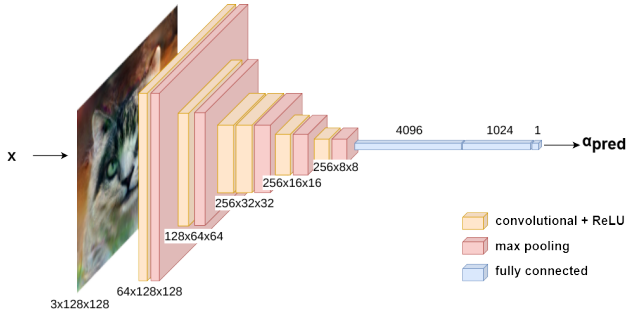


Fig. 5: Modified ACAI/AEAI Critic based on VGG.

The critic is a CNN similar to the encoder, but includes two additional fully connected layers to predict α (see Fig. 5).

2) Loss Functions: The autoencoder loss is denoted as L combining three losses shown below.

$$L(x_i, x_j) = L_R + L_A + L_C \tag{1}$$

$$L_R = \text{LPIPS}(x_i, x_j) \tag{2}$$

$$L_A = \sum_{n=0}^{M} -log(D(\hat{x}(\alpha))) \tag{3}$$

$$L_C = \sum_{n=0}^{M} \|z(\alpha) - \hat{z}(\alpha)\|^2 \tag{4}$$

, where $\alpha = n/M$ is uniformly sampled.

For the critic, we used binary cross-entropy loss as it is the standard loss for discriminators in GANs.

$$L_D = \frac{-1}{N} \sum_{i=1}^{N} [log(D(x)) + log(1 - D(\hat{x}(\alpha)))] \tag{5}$$

, where N is the number of minibatch samples.

D. Training

We trained both models using the cloud compute service Vast.AI [17], as the relatively small amount of compute available in Google Colab was not enough to train the models to convergence. Because of this, we do not have a cell in our notebook that has run our full training procedure; we instead generate our results from downloaded logs and checkpoints. We trained the ACAI model for 1630 epochs and the AEAI model for 766 epochs; we stopped at these points because the loss curves had levelled out and the generated images had qualitatively stopped improving. The cloud compute used an RTX 4070 GPU, and training took 19 hours for the ACAI model and 14 hours for the AEAI model. We did a hyperparameter search for both models where we selected 30 random sets of hyperparameters for each model and trained it for 10 epochs. We then selected the hyperparameters that produced the best qualitative results. The hyperparameters adjusted were the weights of the loss components, the learning rate of the autoencoder and the critic, and the reconstruction loss function (Mean Square Error or LPIPS). The learning rates were randomly sampled from a log uniform distribution from 0.01 to 10^{-5}. The weights of the loss components were similarly randomly sampled from a log uniform distribution from 5×10^{-2} to 1×10^{-3}.

Difficulties in training. GAN-like models are notoriously difficult to train; the two competing models must be kept in equilibrium in order to avoid issues resulting from one of the models dominating the other. A problem we frequently faced due to this was mode collapse; this is when the generator model (in this case our autoencoder) degenerates into outputting the one image which fools the critic the best irregardless of the input images (see Fig. 6). This is counter-intuitively a result of the critic being too effective, as this causes the gradient from the adversarial loss component of the autoencoder to dominate the gradient from other loss components like reconstruction and cycle loss.

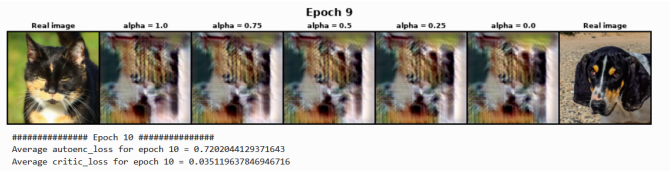


Fig. 6: An example of the outputs produced from an AEAI model trained without weight clipping.

To solve this problem, we utilized weight clipping, a technique introduced in the Wasserstein GAN paper [13]. This technique clamps the weights of the critic so that they are always within a certain small range (in our case, we used $[-0.01, 0.01]$); this creates an upper bound on how effective the optimal critic can be and also prevents the critic from changing too much between each training step (see Fig. 7).

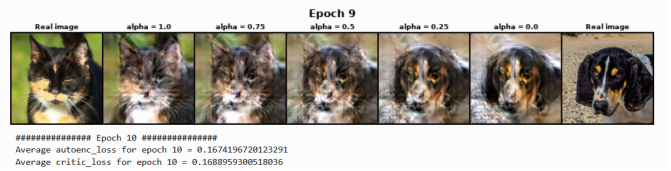


Fig. 7: An example of the outputs produced from an AEAI model trained with weight clipping. Notice that the critic loss is larger than the critic loss without weight clipping.

V. Results

In this section, we evaluate and compare the quality of two models and a simple baseline. Although the quality of interpolations is hard to quantify, the general goal is to produce animal hybrids that combine abstract features of both animals and their backgrounds. To produce believable interpolations, the model has to first learn to encode these high level features, which can include lighting, color, fur patterns, face shape, image segmentation, and pose. Ideally, averaging these high level features will generate an animal hybrid.

An ideal model will likely have good disentanglement between different features of the latent space, as this will

improve the effectiveness of latent space representation and thus the effectiveness of the interpolations. Another important factor is the shape of the latent space the autoencoder learns; the AEAI paper claims that there is significant importance in creating a smooth and locally convex latent space, because by the definition of convexity, linear interpolations in a convex space will remain in the space. The ACAI model mainly focuses on learning this convex latent space, but there are no additional losses to penalize artifacts or other image quality issues that could lead to poor interpolations.

Our baseline is direct linear interpolation (sometimes referred to as a crossfade) in the image space. This does not produce believable animal hybrids, especially in comparison to the other models (see Fig. 8).

For the models, we believe the interpolations qualitatively show that AEAI is better than ACAI for most samples, as it produces sharper and more realistic interpolations (see Fig. 8 and Fig. 10). The authors of the AEAI paper claim this is because of their use of a discriminator, which predicts a probability, rather than a critic which tries to predict α. This is because the task of predicting α is too difficult as the interpolations do not have enough information, which makes the training of the critic ineffective.

We also demonstrate that the AEAI and ACAI models learn to encode some pose information; when tested on small changes in pose, both models seem to learn the correct interpolation between the two poses (see Fig. 10). The AEAI model may be slightly better in this regard, as it seems to produce a less blurry interpolation. However, for large differences in pose, especially with samples of a different species, both the ACAI and AEAI models fail and look similar to crossfade. This suggests that the pose information encoded in the latent space is likely entangled with other features and/or not globally convex.

Quantitative Results. Although it is difficult to quantify the quality of the interpolated images, there are standard metrics typically used for evaluating generative models that we can borrow. We chose to use Fréchet Inception Distance (FID) [16], one of the more commonly used metrics in the literature. Instead of measuring a direct distance between two images, FID measures the difference in distribution between two sets of images. To do this, it first passes the images through a pretrained image classification model (InceptionV3) and takes the second last layer's activations; it uses these values as a proxy for abstract, higher-order image features. It then measures the Fréchet distance between these two sets of activations by making the assumption that they are normally distributed. To quantify our interpolations, we measure the FID between the dataset and the model's interpolations of the dataset with $\alpha = 0.5$; we do this with both the training and validation set to try and reduce the impact of overfitting. The results are shown in Table 1. Overall, the AEAI model performed better than the ACAI model, with a significantly lower FID with both the train and validation sets.

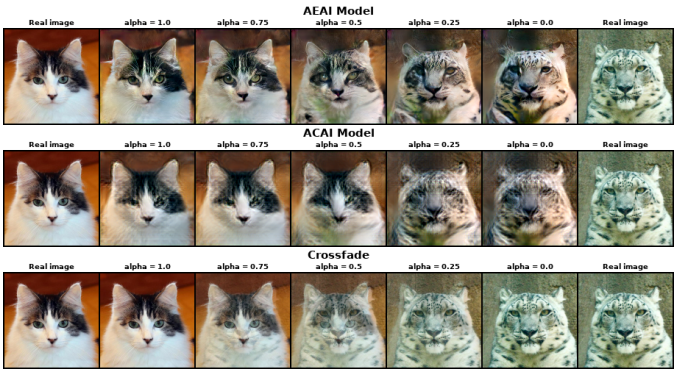


Fig. 8: Interpolation results comparing three strategies. Notably, the ACAI $\alpha = 0.5$ image is similar to the left image, and its eyes are closed. The AEAI model shows improvement with open eyes and combining fur patterns.

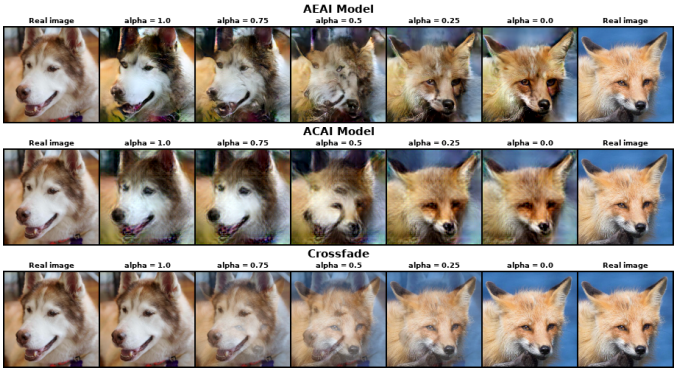


Fig. 9: Interpolation results comparing three strategies when the difference in azimuth angle is large.

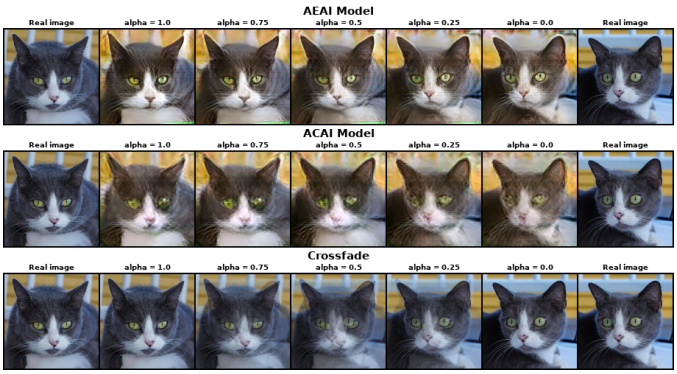


Fig. 10: Interpolation results comparing three strategies for the same cat and small difference in azimuth angle.

	AEAI	ACAI
FID (train, $\alpha = 0.5$)	**49.43**	86.28
FID (val, $\alpha = 0.5$)	**63.39**	100.41

TABLE I: FID scores of interpolations generated by AEAI and ACAI on train and validation set. The better results in each row are bolded (lower is better).

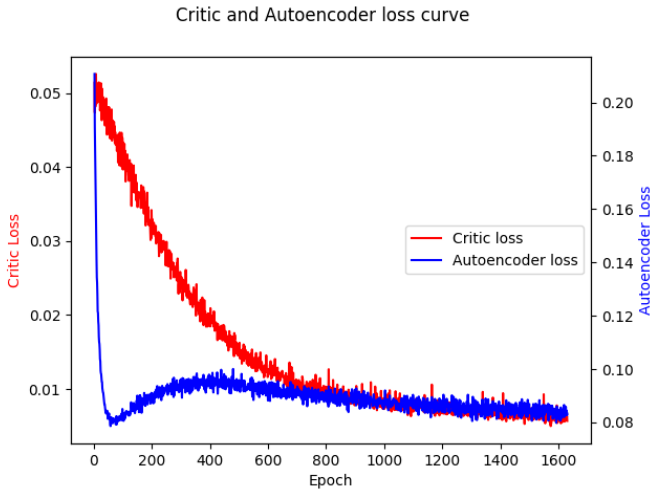


Fig. 11: ACAI loss curve.

Loss Curves. In a GAN type network, the generator and discriminator are competing to get better. As a result, the loss for the generator and discriminator become symmetric, where lowering one loss increases the other. Usually, the generator starts off with high loss because creating realistic images is difficult, while the discriminator starts with low loss, as it is easy to tell apart real from interpolated. After some time, the generator loss improves and the critic loss increases, and they may converge. This pattern is seen in the AEAI model for the first 100 epochs. After that, the critic seems to improve faster than the autoencoder (see Fig. 12 and Fig. 13).

The ACAI loss tells a different story, as the autoencoder has a rapid decline in loss due to a bad critic. As the critic steadily gets better at predicting alpha, it competes with the autoencoder. Eventually, both losses seem to reach equilibrium (see Fig. 11).

VI. Conclusion

While image generation has strong capabilities, the problem of image interpolation is lesser studied, but still has had advancements in recent years. ACAI introduces a relatively simple method using linear interpolation in the latent space and a critic. AEAI improves it by enforcing *admissible interpolations* using additional loss terms. Both these models illustrate that adversarial autoencoders are capable of image interpolation. Intuitively, interpolation requires the autoencoder to learn a high-level understanding of the data, which is demonstrated to be the case in some of our results. In some cases AEAI proves that it has learnt this high-level meaning,

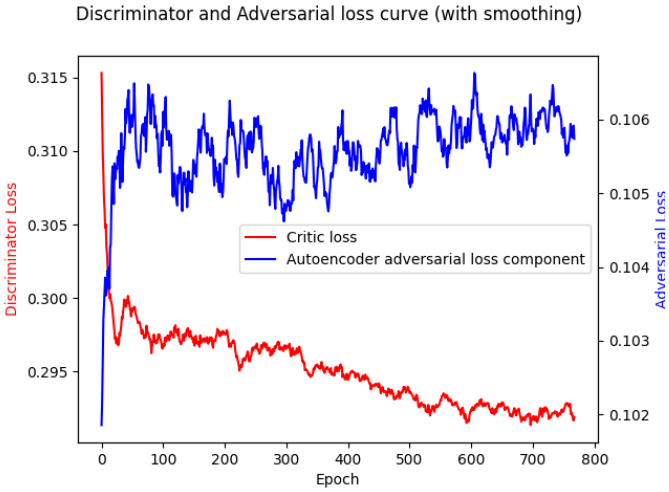


Fig. 12: AEAI critic loss and adversarial loss, smoothed with exponential moving average.

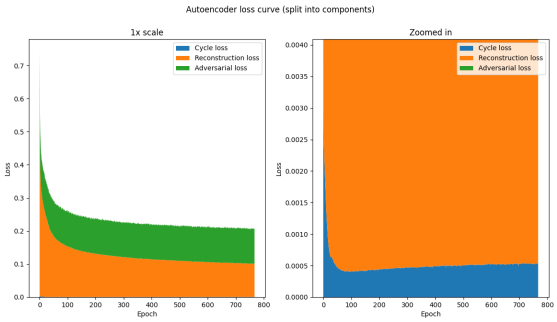


Fig. 13: AEAI Autoencoder loss curve over 766 epochs. (Left) 1x scale loss. (Right) Scaled down loss axis to show cycle loss.

and produces better qualitative and quantitative results than ACAI for animal hybrids. Training GAN's are difficult due to instabilities, mode collapse, and degeneration. Overall, ACAI and AEAI use simple linear interpolation in the latent space, but this is restrictive method, especially for nonconvex spaces. Strategies such as correction terms [3] and crossover interpolation [8] might provide more flexible methods for nonlinear interpolation.

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